Dimensions and determinants of judgements of colour samples and a simulated interior space by architects and non-architects

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Architects and non-architects made Semantic Differential ratings of colour samples (chips) and a simulated interior space (a model). In analyses of the total samples' ratings (architects and non-architects) of (a) colour chips and (b) models, and individual sample analyses, (c) architects' chip judgements, (d) architects' model judgements, (e) non-architects' chip judgements, and (f) non-architects' model judgements, five factors occurred, though not necessarily all in any one analysis. These were: (i) dynamism; (ii) spatial quality; (iii) emotional tone; (iv) evaluation; (v) complexity. Linear correlations between parameters of the Munsell Color System and the above factors in the various analyses were calculated, while parallel analyses were carried out employing a graphical technique described by Sivik (1974*a*) involving isosemantic maps. In all analyses, linear correlations between the colour parameters and judgements were found for the dynamism factor, spatial quality factor, and emotional tone factor. They were associated respectively with chroma, value, and hue. Inspection of the isosemantic maps indicated subsidiary effects of the non-dominant dimensions of a non-linear sort, though the maps also exhibited the linear relations. Linear correlations were low or non-existent for the evaluation and complexity factors, and the complex nature of their determinants was clear from the isosemantic maps. The determinants of judgements were similar for architects and non-architects, with the exception of evaluative judgements for the models in which markedly different determinants were noted.

Comparability of the present findings with other studies carried out in a variety of countries over a 20 year period was high for dynamism, spatial quality, and emotional tone, and it is suggested that there may be something inherent in the response to colour in relation to such judgements. Recent physiological work is discussed, and its limitations in terms of colours sampled and an overconcentration on the hue dimension noted. In contrast, it is suggested that dimensions of judgement, such as evaluation or complexity, reflect to a greater extent culture or training, and are hence independent of the basic colour attributes.

Studies of colour preference and colour connotations span the history of experimental psychology. In recent years, several developments within this area have taken place. First, there has been an increasing tendency to extend the field from studies of colour in its own right to the use of interior (Acking & Küller, 1972) and exterior colour (Sivik, 1974*b*). Both Acking & Küller, and Sivik, indicate the need to undertake studies that attempt in a formal way to relate the effects of colour out of any specific situation to the response to the same colour in an architectural context. The present paper addresses itself to this question by requiring subjects to rate both colours in chip form and in a spatial model in which the same colours were employed.

Second, there has been a growth in interest in the way in which dimensions of judgements concerning colour are related to the coordinates of colour specification systems such as hue, chroma and value. Broadly speaking, three procedures have been employed: (*a*) the fitting of appropriate equations to the data (e.g. Stamm, 1955) or the use of multiple regression (e.g. Wright & Rainwater, 1962); (*b*) a verbal characterization of the relation of dimensions of judgement to the dimensions of colour coordinate systems (e.g. Hogg, 1969); (*c*) graphical plots of judgements in a visual representation of the colour coordinate system (Guilford & Smith, 1959; Sivik, 1974*a*). In the present study, correlation and graphical procedures are both used in order to establish the extent to which the former accurately reflect such relations where correlations do exist, and the extent to which orderly but non-linear relations hold, even in the absence of correlations.

Third, possible contrasts in judgements between those professionally concerned with colour and non-professionals have been suggested (Küller, 1972). In the present study, separate analyses are undertaken for architects and non-architects.

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Method

Subjects

Ten architect subjects and 10 non-architect subjects undertook the rating. The architect subjects were qualified architects and students of architecture. Their mean age was 26 years, and five males and five females took part. The non-architect subjects were all professional people (e.g. librarians, lighting engineers, social workers, etc.). Their mean age was 33 years, and five males and five females took part.

Apparatus

Colour material: For the purposes of this experiment, it was decided to select the colour samples from the new BSI Range (British Standards Institution, 1972, Basic range for the coordination of colours for building purposes, DD17:1972). This range of colours is coordinated on newly defined attributes of hue, greyness, and weight (Building Research Establishment, 1973) and is based on the Munsell Color System with its coordinates of hue (referring to a scale of perceptions ranging from red through yellow, green, blue and back to red), chroma (referring to a scale of perceptions representing a colour's degree of departure from an achromatic colour of the same value), and value (referring to a colour's similarity to one of a series of achromatic colours ranging from light to dark).

Colours were selected from the range in five major hue groups: red (04), yellow (10), green (14), blue (18) and purple (22). They were chosen so that, so far as possible, colours of approximately equal chroma (greyness) and value (weight) were represented for each hue group and so that there was an adequate sampling of colours across the whole range, black and white were also included. Colour chips were 12mm × 17mm. The area of colour in the model interior was the largest sample area available, namely 230mm × 560mm. Forty-four colours were used, all semi-matt. The Munsell reference precedes the BSI reference. They were:

5G 3/4	14 C 39	5Y 4/2	10 B 25
10PB 7/6	22 D 41	7.5R 4.5/16	04 E 53
10PB 4/2	22 B 25	5Y 7/6	10 C 35
5B 8/1	18 B 17	7.5R 7/8	04 D 41
5Y 3/4	10 C 39	N 1.5	00 E 53
10PB 4/10	22 E 53	5Y 8.5/6	10 D 41
10PB 3/8	22 D 45	10R 8/2	04 B 17
5G 3/6	14 D 45	5Y 8/2	10 B 17
5B 6/1	18 B 21	7.5R 3/10	04 D 45
10PB 8/2	22 B 17	5Y 5/8	10 D 45
7.5R 9.25/2	04 C 31	10PB 7/4	22 C 35
5G 9/1	14 C 31	7.5B 7/5	18 D 41
6.25Y 8.5/13	10 E 53	5G 7/2	14 C 35
5B 4/1	18 B 25	7.5R 7/6	04 C 35
5G 5/10	14 E 53	10R 6/2	04 B 21
5G 7/4	14 D 41	10PB 3/6	22 C 39
10R 4/2	04 B 25	7.5B 3/6	18 D 45
10PB 9/2	22 C 31	7.5B 7/3	18 C 35
7.5R 3/6	04 C 39	5Y 6/2	10 B 21
N 9.5	00 E 53	5B 9.25/1	18 C 31
10PB 6/2	22 B 21	10B 4/10	18 E 53
5Y 9/2	10 C 31	7.5B 3/4	18 C 39

Presentation of material: The large panels of colour were presented in a model of a room space (560 mm×280 mm×900 mm), the coloured panel representing the end wall. The floor and side walls were painted grey (Munsell N7; 00 A 05) and the ceiling was covered with a fine white gauze through which the light was transmitted into the model. When each coloured panel was removed, the end wall was of the same grey. The panels were viewed at a distance of 1000 mm and the colour chips were also presented in the model at a distance of 600 mm.

The laboratory was illuminated by six pairs of 65/80 W artificial daylight tubes, giving a level of illumination of 260 lux. A small, artificial-daylight tube was suspended over the model producing a luminance level of 30 candelas/m^2 on both the end wall of the model and the colour chip. The reflectance of the grey surface was 42 per cent. The colour rendering index of the daylight tube was 95.

Procedure

Subjects were tested individually. They reported their judgements verbally, qualifying the adjectives with the words 'neutral', 'slightly', 'moderately', or 'extremely'. These were recorded by the second author. Colour chips and models were rated in separate sessions. Instructions for the rating of the colour chips were similar with the exception that the subjects were asked to make judgements of 'the effect of the colour' rather than the 'effect of the space'. Scales were chosen to incorporate those in Hogg (1969), supplemented by additional adjectives relevant to spatial judgements. Twenty-four scales were used, their order of presentation being counterbalanced with regard to adjective position and scale order: 1. exciting–calming; 2. open–closed; 3. cold–hot; 4. dynamic–static; 5. hard–soft; 6. vibrant–still; 7. austere–lush; 8. modern–traditional; 9. unusual–usual; 10. complex–simple; 11. pleasant–unpleasant; 12. fresh–stale; 13. weak–strong; 14. cramped–spacious; 15. blatant–muted; 16. controlled–uncontrolled; 17. obvious–subtle; 18. free–constricted; 19. active–passive; 20. introverted–extraverted; 21. dull–sharp; 22. private–public; 23. receptive–repellent; 24. loose–tight.

Results

The mean rating of each colour chip and each model on each scale was calculated for (*a*) all subjects' colour chip ratings (i.e. architects and non-architects), (*b*) all subjects' model ratings, (*c*) architects' colour chip ratings, (*d*) architects' model ratings, (*e*) non-architects' colour chip ratings, (*f*) non-architects' model ratings. Each of the resulting 24×44 matrices were analysed separately by correlating ratings across scales to produce 24×24 product moment correlation matrices. These were subjected to principal components analysis, followed by varimax rotation. Table 1 presents the factors identified, and their associated percentage variance (Table 1).

In addition, the product moment correlations between the loadings on each of the four factors across subsamples were computed (Table 2).

In order to establish the relation of the factor dimensions to hue, value, and chroma, the factor loadings were correlated with quantifications of these attributes for colour chips and

Table 1. Percentage of total variance accounted for by factors accounting for 4 per cent or more of the total variance

	Colour chips			Models		
Factors identified	Full sample	Architects	Non-architects	Full sample	Architects	Non-architects
Dynamism	46·5 (I)	42·1 (I)	45·6 (I)	54·7 (I)	47·7 (I)	55·7 (I)
Spatial quality	29·9 (II)	29·1 (II)	28·3 (II)	21·8 (II)	21·5 (II)	21·9 (II)
Emotional tone	4·5 (IV)	10·0 (III)	10·1 (III)	8·9 (III)	10·7 (III)	7·6 (III)
Complexity	9·7 (III)	—	4·9 (IV)	—	—	—
Evaluation	—	5·7 (IV)	—	5·4 (IV)	6·6 (IV)	4·1 (IV)

Table 2. Product moment correlations between factor scores for scales on each factor for architects and non-architects

		Colour sample			
		Chips		Models	
Subject group	Factor	Architects	Non-architects	Architects	Non-architects
Chips	I – Dynamism	—	0·79	0·96	0·87
Architects	II – Spatial quality	—	0·90	0·95	0·93
	III – Emotional tone	—	0·64	0·85	0·78
	IV – Evaluation	—	0·21	0·78	0·00
Non-architects	I – Dynamism		—	0·79	0·70
	II – Spatial quality		—	0·89	0·89
	III – Emotional tone		—	0·53	0·62
	IV – Complexity		—	0·18	0·12
Models	I – Dynamism			—	0·91
Architects	II – Spatial quality			—	0·92
	III – Emotional tone			—	0·65
	IV – Evaluation			—	0·03
Non-architects	I – Dynamism				—
	II – Spatial quality				—
	III – Emotional tone				—
	IV – Evaluation				—

Table 3. Product moment correlations between weightings of factors I–IV of colour chip and model analyses for architects with hue, chroma, and value for total, sample, architects, and non-architects

Chips (total sample)	Hue	Value	Chroma
Factor			
I – Dynamism	0·02	0·12	−0·88***
II – Spatial quality	−0·06	−0·89***	0·19
III – Complexity	−0·22	0·34*	−0·17
IV – Emotional tone	−0·74***	−0·01	0·33*
Models (total sample)			
Factor			
I – Dynamism	−0·04	0·15	−0·89***
II – Spatial quality	0·03	−0·90***	−0·26
III – Emotional tone	−0·67***	0·06	0·29
IV – Evaluation	0·08	0·20	−0·11
Chips (architects)			
Factor			
I – Dynamism	0·02	0·07	−0·86***
II – Spatial quality	0·01	−0·91***	0·24
III – Emotional tone	−0·61***	−0·09	0·33*
IV – Evaluation	−0·01	0·24	−0·18
Models (architects)			

Table 3. cont.

Chips (total sample)	Hue	Value	Chroma
Factor			
I – Dynamism	0·00	0·08	−0·87***
II – Spatial quality	0·07	−0·92***	0·31*
III – Emotional tone	−0·57***	0·04	0·24
IV – Evaluation	−0·12	0·28	−0·06
Chips (non-architects)			
Factor			
I – Dynamism	0·33*	0·13	−0·91***
II – Spatial quality	−0·16	−0·86***	0·14
III – Emotional tone	−0·39*	0·18	−0·30*
IV – Complexity	−0·38*	0·34*	−0·14
Models (non-architects)			
Factor			
I – Dynamism	−0·05	0·23	−0·88***
II – Spatial quality	0·05	−0·89***	0·16
III – Emotional tone	−0·62***	−0·05	0·46***
IV – Evaluation	−0·05	0·10	−0·09

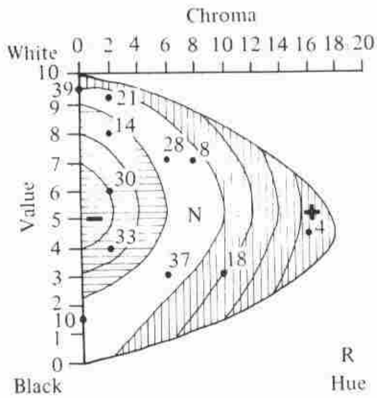
* $P < 0{\cdot}05$ (d.f. = 40, two-tailed test); ** $P < 0{\cdot}01$ (d.f. = 40, two-tailed test); *** $P < 0{\cdot}001$ (d.f. = 40, two tailed test).

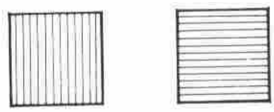
models for both full and subsamples. The hues were quantified by their clockwise angular displacement in the Munsell Hue circle from 5R, which was taken as 0°. Value and chroma were quantified as in the Munsell notation. Black and white were not included in the analysis. The correlations appear in Table 3.

Isosemantic maps, enabling graphical illustrations on sections through the colour solid, were drawn. In order to give an indication of the theoretical shape of the colour solid, a line was drawn encompassing the colour chips shown in the *Munsell Book of Color* (1963). The section was drawn through the colour solid at the sector containing the greatest number of hue samples, and all the colour chips from a BSI hue group were located on that triangle (Fig. 1).

The scores for each colour sample on each factor were plotted on the maps, and isosemantic lines, i.e. lines of equal score, were interpolated at intervals of 0·5. The neutral zone was defined between +0·5 and −0·5, and was left unshaded. Areas with positive scores were shaded vertically, and areas with negative scores, horizontally. Where isosemantic lines were interpolated across the vertical axis, they were shown as a broken line. Shading codes for specific factors also appear in Fig. 1. Maps were drawn for all six analyses, but are here represented by those for the full sample colour chip analysis (Fig. 2) and one factor from the full sample model analysis (Fig. 3).

The statistical and graphical analyses were carried out independently by two of the authors in order to ensure that the placement of isosemantic contours was not informed by the correlational analysis.





Dynamism	Active Sharp Exciting Etc.	Passive Dull Calming Etc.
Spatial quality	Closed Constricted Cramped Etc.	Open Free Spacious Etc.
Emotional tone	Hot Soft Lush Etc.	Cold Hard Austere Etc.
Complexity	Complex Unusual Etc.	Simple Usual Etc.
Evaluation	Repellent Unpleasant Etc.	Receptive Pleasant Etc.

Figure 1. Illustrative isosemantic map: Horizontal shading indicates negative attribute; white indicates neutrality (defined by weightings of $\pm 0{\cdot}5$); vertical shading indicates positive attribute. Contours define differences of $0{\cdot}5$ in weightings.

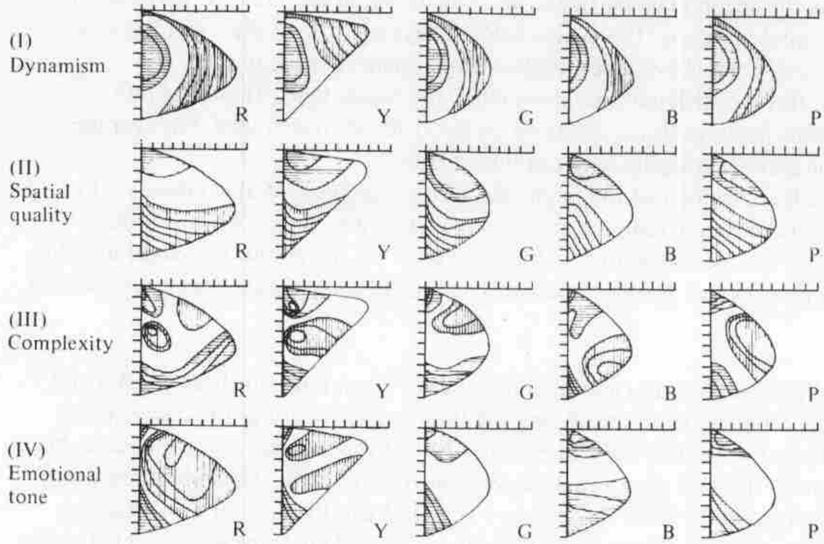


Figure 2. Isosemantic maps for factors I–IV for colour chips (total sample).

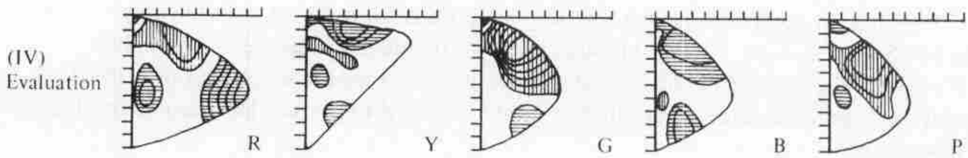


Figure 3. Isosemantic maps for the evaluative factor for models (total sample).

Discussion

Dimensions and their determinants

Five factors accounting for more than 4 per cent of the total variance occurred in the six principal components analyses. These may be characterized as dynamism, spatial quality, emotional tone, complexity, and evaluation.

Dynamism. This was the largest factor in all analyses, accounting for between 42·1 and 55·7 per cent of the total variance. The scales loading most highly on this factor were: exciting–calming, dynamic–static, hard–soft, vibrant–still, austere–lush, modern–traditional, fresh–stale, weak–strong, blatant–muted, obvious–subtle, active–passive, introverted–extraverted, and dull–sharp. For the full group analyses for both chips and models, all these scales had loadings beyond $\pm 0{\cdot}5$. The correlation (Table 2) between the factor loadings of the 24 scales for the various subject groups are all highly significant. The correlation between factor scores for models and chip loadings for architects was $r = 0{\cdot}96$, in contrast with $r = 0{\cdot}70$ for non-architects. Agreement between architects and non-architects is higher for models than for colour chips: $r = 0{\cdot}91$, as compared with $r = 0{\cdot}79$.

In the main sample analysis, the main correlate of dynamism was chroma. For chips, the correlation between factor scores and degree of chroma was $r = -0{\cdot}88$ (d.f. = 40, $P < 0{\cdot}001$), and for models was $r = -0{\cdot}89$ (d.f. = 40, $P < 0{\cdot}001$). The relation was such that as the chroma increased it was judged as being more dynamic. Neither hue nor value made a linear contribution to this effect.

Spatial quality. This was the second largest factor in all analyses, accounting for between 21·5 and 29·9 per cent of the total variance. The scales loading beyond ±0·5 in the two total sample analyses were: open–closed, complex–simple, weak–strong, cramped–spacious, uncontrolled–controlled, free–constricted, private–public, and loose–tight. Table 2 shows correlations between factor loadings of all scales on factor II for the subgroups. On average, these are higher than for factor I, ranging between 0·89 and 0·95.

In the main sample analysis, value had the main effect on judgements of spaciousness. For chips, the correlation between factor scores and degree of value was $r = -0{\cdot}89$ (d.f. = 40, $P < 0{\cdot}001$), and for models was $r = -0{\cdot}90$ (d.f. = 40, $P < 0{\cdot}001$). As the colour increased in value, so it was considered more spacious. Neither hue nor value made a significant linear contribution to this dimension.

Emotional tone. All analyses yielded an emotional tone factor accounting for between 4.5 and 10·7 per cent of the total variance, and constituting the third factor in all but the colour chips full sample analysis. The scales loading beyond ±0·5 in the full sample analyses were: cold–hot, hard–soft, and austere–lush. Table 2 shows that correlations between the subsamples are again significant, though on average lower than for factors I and II. As in the case of those factors, the highest correlation was between the two sets of loadings for colour chips and models for the architects.

In the main sample analysis, hue was the main linear correlate of factor IV judgements. For chips, the correlation was $r = -0{\cdot}74$ (d.f. = 40, $P < 0{\cdot}001$), and for models was $r = -0{\cdot}67$ (d.f. = 40, $P < 0{\cdot}001$). In addition, chroma was also significantly correlated with the chip judgements ($r = 0{\cdot}33$, d.f. = 40, $P < 0{\cdot}05$) though not with the model judgements ($r = 0{\cdot}29$, d.f. = 40, $P = 0{\cdot}06$). The relations indicate that the colours decreased in emotional tone (became cooler, harder, and more austere) round the hue circle from red to purple. Decreasing chroma enhanced this effect.

Complexity. This occurs as factor III in the full sample colour chips analysis (9·7 per cent of total variance) and factor IV in the non-architects' colour chips analysis (4·9 per cent of total variance). In the full sample analysis, the following scales have loadings beyond ±0·5: unusual–usual, complex–simple, private–public, and modern–traditional. In the full sample analysis, value was the main linear correlate of judgements of complexity ($r = 0{\cdot}34$, d.f. = 40, $P < 0{\cdot}05$). Hue did not correlate significantly with complexity ($r = -0{\cdot}22$, d.f. = 40, $P > 0{\cdot}05$). These correlations reflect a tendency for complexity to decrease as the colours become of lower value and to increase with progression round the hue circle from red to purple. (Black and white were also regarded as simple 'colours'.) Similar relations hold with the non-architects' colour chip judgements. Value and complexity correlated $r = 0{\cdot}34$ (d.f. = 40, $P < 0{\cdot}05$) while hue and complexity correlated $r = -0{\cdot}38$ (d.f. = 40, $P < 0{\cdot}05$).

Evaluation. This occurs as the fourth factor in all model analyses and in the architects' colour chip analysis, and accounts for between 4·1 and 6·6 per cent of the total variance. The main scales associated with the evaluation factor are pleasant–unpleasant and receptive–repellent. Table 2 shows that there was a high correlation between the loadings for the colour chip and model colours for architects of 0·78. However, the correlation between the architects and non-architects for models was only $r = 0{\cdot}03$.

There were no significant correlations between hue, value and chroma and the factor loadings for any of the analyses.

Selected isosemantic maps comparison

The relations between the dimensions described above and the coordinates vary from strong linear correlations in the case of dynamism and spatial quality, through to somewhat weaker, multiply determined relations in the case of emotional tone, and an absence of such relations in the case of evaluation. The characteristics of these relations may be explored further through an illustration of the same data in the form of isosemantic maps. This will be limited to the full sample chip analysis, with its factors of dynamism, spatial quality, complexity, and emotional tone, and the evaluative factor from the full sample model analysis. The maps appear in Figs 2 and 3.

Dynamism (*colour chips*). The relation between chroma and dynamism indicated by the correlation coefficient is illustrated in the isosemantic maps. The predominant direction of the contours is vertical, with a progressive shift from negative to positive value as chroma increases on the horizontal axis. In addition, an effect of value may be discerned, but does not bear a linear relation to the weightings. The bowed effect of the contours, particularly clear for the red, green, blue and purple maps, indicates that high or low value may compensate for low chroma. Thus, as chroma is reduced, an increase or decrease in value from the midpoint of the value scale increases the degree of dynamism. Only in the yellow map does this not hold entirely.

Spatial quality (*colour chips*). The relation between chroma and spatial quality indicated by the correlation coefficient is illustrated in the isosemantic maps. Here, the contours tend to be horizontal. There is a progressive increase in spaciousness as we move vertically through the map. A slight effect of chroma of a non-linear nature may also be noted, though the form that this takes depends upon hue. For the green, blue, and purple maps, increasing chroma generally leads to an increase in spaciousness. For yellow, the relation of spaciousness to chroma is curvilinear. As chroma increases from the value axis, spaciousness increases, then decreases. For red, the effect of chroma is similar to that of yellow. The effect of the bowing of the contours appears to be related to the value of the most saturated colour samples, as the neutral area tends to extend from between the values of six and eight on the value axis to the zone of maximum chroma.

Complexity (*colour chips*). Limited relations with value and hue were found in the correlational analyses. Similarly, the isosemantic maps do not present as clear a picture as for the preceding factors. Systematic effects of value can be discerned only in limited regions of the maps, e.g. the lower half of the green map. The hue effect is also small, as there is only a slight increase in the vertically shaded area, indicative of an increase in complexity from the red map through to the purple map. The main impression given by the maps is of an interaction between value and not hue, but chroma. Colours of low chroma, but of low or high value, are judged as simple. This holds for yellow, blue, and purple, and to a lesser extent for red and green. Reds and greens of low value are consistently simple across the chroma range. Complex colours are generally of medium value.

Emotional tone (*colour chips*). Hue, predominantly, but also chroma, contributes to the emotional tone of the colour. The influence of hue is clearly seen in the isosemantic maps. As we move from the red map, the majority of which indicates high emotional tone, to the purple map, so the degree of this quality decreases, reaching a minimum at the blue map. In the case of the high emotional tone hues, red and yellow, only colours of low chroma and high or low value are of negative emotional tone. In the case of green, blue, and purple, the maps indicate entirely low to neutral emotional tone. The green map denotes greatest neutrality with respect to this factor. The significant correlation between chroma and the factor reflects the tendency for low chroma to be associated with low emotional tone and high chroma with high emotional tone or neutral quality. Only the blue map indicates any deviation from this.

Evaluation (*models*). The reason for the absence of significant linear correlations is immediately apparent from the related isosemantic maps (Fig. 3). The Yellow and Blue maps both indicate that these hues are in the main preferred, or judged as neutral. In contrast, the green and purple maps show predominantly low preference or neutrality. Red lies between the two groups, showing almost equal areas of high and low preference. Within all maps, however, deviations from these generalisations may be discerned that are dependent upon interactions of the hue with value and chroma, e.g. high values of red in medium and low chromas are predominantly unpleasant.

Two general conclusions may be drawn from these comparisons between the correlation coefficients and the isosemantic maps. First, the presence or absence of a strong linear correlation between a colour coordinate and a dimension of judgement is directly reflected in the isosemantic maps. Second, even where such correlations have been found, subsidiary effects do emerge in the maps showing subtle interactions between the dimensions. Where no correlation has been detected, complex interactions between the colour dimensions can be found in the maps which determine the judgements.

Comparability of factor structures and determinants

It is clear that there is a high degree of consistency in the factor patterns resulting from the architect and non-architect group ratings. This holds strongly for the two main dimensions, dynamism and spatial quality, and to a lesser extent for factor III, emotional tone. However, in judgements of colour chips and models by non-architects, there is a marked difference from the architects in factor IV. The main contribution to the differences for colour chips comes from the loading of pleasant–unpleasant (0·73) and receptive–repellent (0·71) on factor I for the non-architect group in contrast to the high loading of these scales on factor IV for the architect group. It is possible that for architects, preference for colour is separated from the specifically dynamic connotation of the colour and independent judgements are made. No such separation is made by non-architects, though they are able to make such a separation when an extrinsic context is provided by the model. Even here pleasantness has a relatively high loading on the dynamism factor (0·60), in contrast to that of the architects (0·28).

As may be seen from Table 3, the correlation of dynamism, spatial quality, and emotional tone with the three colour coordinates was broadly the same for both architects and non-architects. Inspection of the isosemantic maps for the groups confirmed this, and also indicated that some of the subsidiary contributions noted in the above discussion of the full sample maps held for the subsamples. For the architects' colour chip and model maps, strong similarities exist in the determinants of evaluation. However, when the preferences of architects and non-architects for models are compared, striking differences in which colours are preferred and not preferred emerged. In addition, inspection of the factor structure of the two groups indicated that whereas the architects associated 'pleasant' and 'receptive' with 'modern' and 'controlled', the non-architects associated these judgements with 'fresh' and 'lush'.

Comparability of dimensions and determinants with other studies

The present study offers a close replication of Hogg (1969). There, four factors occurred in both the rating of single colours and their combination, obtrusiveness (here called 'dynamism'), emotional tone, preference (evaluation), and usualness (complexity). All these factors were represented at some point in the present analyses with the addition of a spatial quality factor.

This represented the additional scales related to spatial characteristics. The comparability of the factors extends to the determinants of dynamism and emotional tone in the present and the earlier work. While hue was noted as a determinant of evaluation in both studies, the isosemantic maps presented here indicate a more complex and interactive situation than could be suggested by the earlier analysis. In both studies, hue and value were suggested as influences of judgement of complexity.

While variations in choice of scales limits the directness with which the present work and that of other authors can be compared, a collation of the factors arising from different studies does suggest marked similarities. Thus, work on single colours typically yields factors comparable to our dynamism, emotional tone, and evaluation (Wright & Rainwater, 1962; Sivik, 1974*a*), as does work on two-colour (Hogg, 1969) and three-colour combinations (Nayatani *et al.* 1969). When complex realistic stimuli are considered, this generalization still tends to hold and extends to the occurrence of a spatial factor, as may be seen in the work of Acking & Küller (1972) on actual and illustrated interiors and exteriors of buildings, Sivik (1974*b*) on exterior colour, and Inui (1966) on actual and illustrated interiors.

The determinants of colour judgements between the present and other studies also bear comparison. This holds particularly for the relation of dynamism and chroma, and emotional tone and hue (e.g. Wright & Rainwater, 1962). Where a spatial factor has occurred, e.g. Acking & Küller (1972), value is also seen as the main determinant. Typically, evaluation does not have any obvious linear correlates. A visual comparison of Sivik's (1974*a*) isosemantic maps for dimensions comparable to those described here also suggests a high degree of similarity in some instances, notably for our dynamism and his 'excitement' factor.

The similarity of findings noted is impressive since the work described has been carried out over a period of nearly 20 years in the United Kingdom, Germany, Sweden and Japan. The most striking point of comparison relates to the influence of chroma on dynamism and like factors, hue on emotional tone, and possibly value on spatial quality. However, for the recurrent dimension of evaluation or preference, differences emerge in the determinants. Oyama *et al.* (1966) contrasted the colour judgements of Japanese and United States individuals and while finding agreement on activity and potency factors, did not find such high agreement in relation to evaluation. When differing professional groups are contrasted, as in the present work and that carried out by Küller (1972) on architects, marked differences in colour preference occur even when other dimensions and their determinants are very similar.

These results suggest that it may be possible that where determinants of judgement reflect fundamental dimensions of colour such as hue, chroma and value, there is some inherent, possibly biological, relation between colour and its connotations as expressed in language. Where no such linear relation exists, then judgements may display to a greater extent the influence of learning. Unfortunately, studies of biological response to colour and the extent to which associative or learning factors influence our response to colour are not available in the form that would allow us to make further inferences on this hypothesis. While Goldstein (1942) set a keynote in suggesting that colour influenced both central and autonomic nervous system responding, studies have typically limited themselves to a small number of colours, varying only in hue, and have not provided clear support for Goldstein's position (e.g. Goodfellow & Smith, 1973; Jacobs & Hustmyer, 1974). Similarly, studies of learning are few and have limited themselves to demonstrations of the influence of both operant and classical conditioning effects on response to colour (Eisman, 1955; Busch & Evans, 1977). It is clear that if the effects of colour dimensions on judgements of colour connotations and evaluation are to advance, both biological and learning studies must expand the dimensions of colour explored and the judgemental dimensions considered.

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A detailed report of the work described in this paper is available from the first author. It is necessary to charge for the cost of Xeroxing this material.

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